INCH-POUND

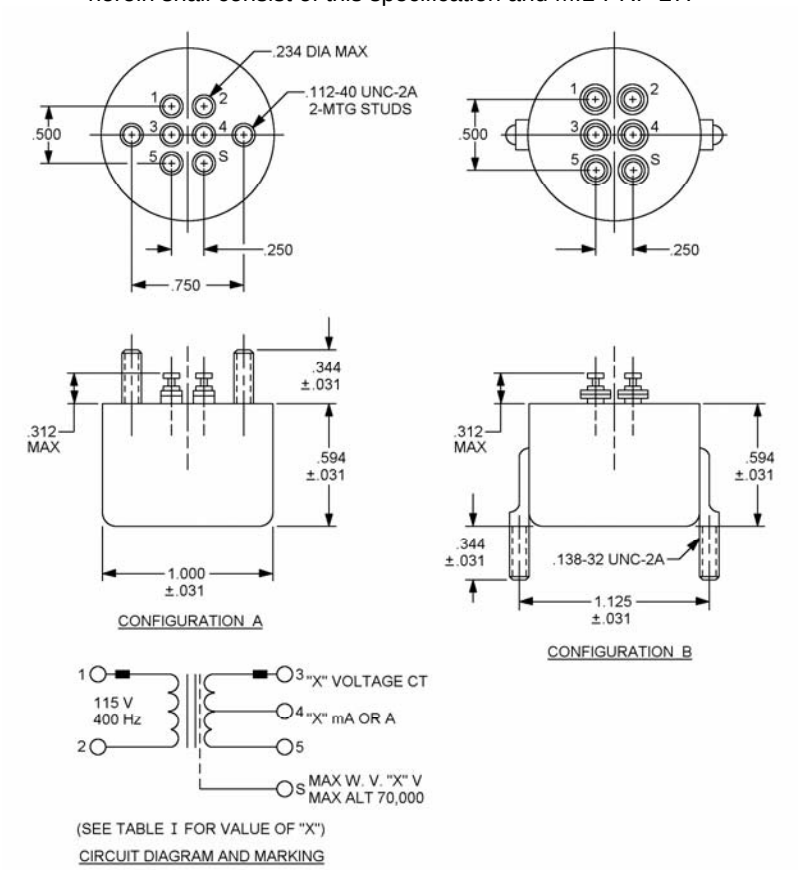
MIL-PRF-27/49D
6 May 2005
SUPERSEDING
MIL-PRF-27/49C
7 February 1984

PERFORMANCE SPECIFICATION SHEET

# TRANSFORMERS, POWER, 1.5 VOLT AMPERES, 400 HZ, ELECTROSTATIC SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformers described herein shall consist of this specification and MIL-PRF-27.




| Inches | mm |
|---|---|
| .031 | 0.79 |
| .112 | 2.84 |
| .138 | 3.51 |
| .234 | 5.94 |
| .250 | 6.35 |
| .312 | 7.92 |
| .344 | 8.74 |
| .500 | 12.70 |
| .750 | 19.05 |
| .812 | 20.62 |
| 1.000 | 25.40 |
| 1.125 | 28.58 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the side of the case.
5. Terminals are color coded for permanent identification in addition to terminal marking shown. Ground terminal "S" is black.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.



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REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 1.25 ounces, maximum.

Duty cycle: Continuous.

Case: Metal enclosed.

Material: Steel.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Feed thru.

Terminal height: .312 inch (7.92 mm), maximum.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 5 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 3.0 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 70 percent minimum at rated load.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{Voltage\ (no\ load) - Voltage\ (rated\ load)}{Voltage\ (rated\ load)} \quad x \quad 100$

Shall not exceed 18 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 17°C, maximum with 115 volts, 400 Hz across (1-2) at ambient temperature of 113°C.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.



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VERIFICATION:

Extent of qualification:

Qualification testing and approval to M27/49-44 A or B, M27/51-44 A or B, M27/53-44 A or B, M27/56-44 and M27/66-27 shall be sufficient to grant qualification approval to MIL-PRF-27/48 through MIL-PRF-27/66 inclusive, all parts.

Qualification testing and approval to M27/49-44 shall be sufficient to grant qualification approval to M27/49-01 through M27/49-44.

Part or Identifying Number (PIN), example:

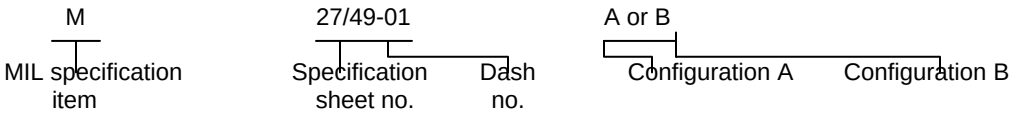



TABLE I. Electrical ratings.

| Dash number | Secondary voltage (3-5) (V rms) ±3% | Secondary current (3-5) (amps rms) | Working voltage (V rms max) |
|---|---|---|---|
| 01 A or B | 6 | 0.250 | 325 |
| 02 A or B | 6.3 | 0.238 | 325 |
| 03 A or B | 7 | 0.214 | 325 |
| 04 A or B | 7.5 | 0.200 | 325 |
| 05 A or B | 8 | 0.188 | 325 |
| 06 A or B | 9 | 0.167 | 325 |
| 07 A or B | 10 | 0.150 | 325 |
| 08 A or B | 11 | 0.136 | 325 |
| 09 A or B | 12 | 0.125 | 325 |
| 10 A or B | 12.6 | 0.119 | 325 |
| 11 A or B | 13 | 0.115 | 325 |
| 12 A or B | 15 | 0.100 | 325 |
| 13 A or B | 16 | 0.094 | 325 |
| 14 A or B | 18 | 0.083 | 325 |
| 15 A or B | 20 | 0.075 | 325 |
| 16 A or B | 22 | 0.068 | 325 |
| 17 A or B | 24 | 0.063 | 325 |
| 18 A or B | 26 | 0.058 | 325 |
| 19 A or B | 28 | 0.053 | 325 |
| 20 A or B | 30 | 0.050 | 325 |
| 21 A or B | 33 | 0.045 | 325 |
| 22 A or B | 36 | 0.042 | 325 |
| 23 A or B | 39 | 0.038 | 325 |
| 24 A or B | 43 | 0.035 | 325 |
| 25 A or B | 47 | 0.032 | 325 |
| 26 A or B | 51 | 0.029 | 325 |
| 27 A or B | 56 | 0.027 | 325 |
| 28 A or B | 62 | 0.024 | 325 |
| 29 A or B | 68 | 0.022 | 325 |
| 30 A or B | 75 | 0.020 | 325 |
| 31 A or B | 82 | 0.018 | 325 |
| 32 A or B | 91 | 0.016 | 325 |
| 33 A or B | 100 | 0.015 | 325 |
| 34 A or B | 110 | 0.014 | 325 |
| 35 A or B | 115 | 0.013 | 325 |
| 36 A or B | 120 | 0.0125 | 325 |
| 37 A or B | 130 | 0.0115 | 325 |
| 38 A or B | 150 | 0.010 | 325 |
| 39 A or B | 180 | 0.0085 | 325 |
| 40 A or B | 200 | 0.0075 | 440 |
| 41 A or B | 220 | 0.007 | 440 |
| 42 A or B | 240 | 0.006 | 440 |
| 43 A or B | 270 | 0.0055 | 580 |
| 44 A or B | 300 | 0.005 | 580 |



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Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

The margins of this specification are marked with vertical lines to indicate where changes from the previous Issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army – CR
Navy – EC
Air Force – 11

Preparing activity:
DLA - CC

(Project 5950-1192)

Review activities:
Army – CR4, MI
Navy – AS, MC, OS
Air Force – 19, 99

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